FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

HALF COURSE TEST – VIII

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

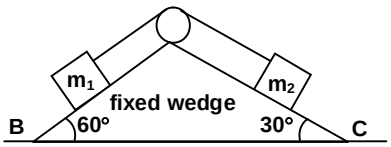
SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

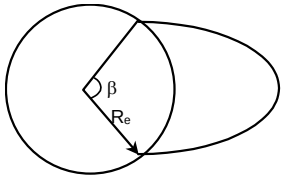
1. Ideal gas is in the container with a piston as one of the walls. There is a little hole in the wall opposite to piston, through which gas leaks out of the container at a rate R (number of molecules escaping the container per second). Now the gas is heated as well as compressed such that its temperature becomes 4 times its previous value, and pressure becomes 8 times its previous value. Now the rate of the gas out flow will be:
(A) 2R (B) 4R
(C) 8R (D) 16R

2. Two small masses m_1 and m_2 are at rest on a frictionless, fixed triangular wedge whose angles are 30° and 60° as shown. They are connected by a light inextensible string. The side BC of wedge is horizontal and both the masses are 1 metre vertically above the horizontal side BC of wedge. There is no friction between the wedge and both the masses. If the string is cut, which mass reaches the bottom of the wedge first? (Take g = 10m/s^2)



(A) Mass m_1 reaches the bottom of the wedge first.
(B) Mass m_2 reaches the bottom of the wedge first.
(C) Both reach the bottom of the wedge at the same time.
(D) It's impossible to determine from the given information.

3. A particle is projected from the surface of the earth (of radius R_e and mass M_e) with speed equal to $\sqrt{\frac{GM_e}{R_e}}$, at certain angle from local horizontal as shown in the figure such that the angle subtended by arc between launching & landing site at earth's centre is $\beta = 90°$. Calculate the maximum separation of the particle from the centre of the earth. Consider earth to be uniformly dense and air resistance to be absent.

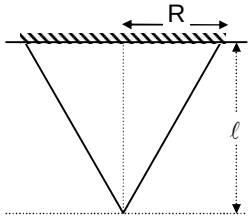


(A) $R_e\sqrt{2}$ (B) $R_e(\sqrt{2}+1)$
(C) $R_e\frac{(\sqrt{2}+1)}{\sqrt{2}}$ (D) $R_e\frac{(\sqrt{2}+1)}{(\sqrt{2}-1)}$

Space for rough work

4. A uniform circular disc of radius r undergoes small oscillations about a horizontal axis which lies in its own plane. Find distance of axis from centre for the period to be minimum.
(A) r/2 (B) r/4
(C) 3r/4 (D) 3r/8

5. A cone of circular cross–section having base radius R, height ℓ and mass m is suspended from its base as shown in the figure. The material of cone has Young's modulus Y. The acceleration due to gravity is g. Find elastic potential energy stored in the cone.



(A) $\frac{m^2g^2\ell}{10\pi R^2Y}$ (B) $\frac{m^2g^2}{5\pi R^2Y}$
(C) $\frac{m^2g^2\ell}{2\pi R^2Y}$ (D) $\frac{m^2g^2\ell^2}{10\pi R^2Y}$

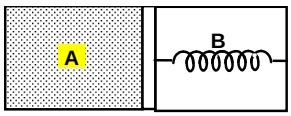
6. A particle of mass m is constrained to move along x-axis. The only force acting on the particle is given by $F(x)=F_0\cos\left(2\pi\frac{x}{\lambda}\right)$, where x is position of the particle on the x-axis. It is at rest on one of the equilibrium position but the equilibrium is unstable. Due to slight disturbance it moves, write the speed of the particle when it has covered a displacement of magnitude L.
(A) $\sqrt{\frac{F_0\lambda}{m\pi}\cos\left(\frac{2\pi L}{\lambda}\right)}$ (B) $\sqrt{\frac{F_0\lambda}{m\pi}\left(1-\cos\frac{2\pi L}{\lambda}\right)}$
(C) $\sqrt{\frac{F_0\lambda}{m\pi}\sin\left(\frac{2\pi L}{\lambda}\right)}$ (D) $\sqrt{\frac{F_0\lambda}{m\pi}\left(1-\sin\frac{2\pi L}{\lambda}\right)}$

Space for rough work

Multiple Correct Choice Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant K is connected with the piston and the wall of the container as shown. Initially the spring is unstretched. The ideal gas in chamber A is allowed to expand. Let in equilibrium the spring be compressed by a length x_o, then



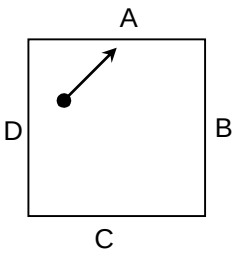
(A) Final pressure of the gas is Kx_0/S

(B) Work done by the gas is $\frac{1}{2} kx_0^2$

(C) Magnitude of change in internal energy of the gas is $\frac{1}{2} kx_0^2$

(D) none of these

8. A smooth uniform square wall frame of mass m of edge length 2a and small height lies in a smooth horizontal plane. At $t = 0$ a particle of mass m hits with speed $v\sqrt{2}$, one of the inner wall of the frame horizontally at an angle 45^0 from wall normal. Collision is perfectly elastic. The initial collision takes place at the mid point of wall A.



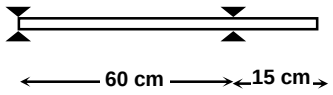
(A) Next collision takes place when $t = \frac{a}{v}$

(B) KE of frame when $t = \frac{3a}{2v}$, is mv^2

(C) after collision at B next collision takes place at wall D

(D) particle and frame comes to rest at regular intervals of time

9. A metal rod of length 75 cm is clamped at two points as shown in figure. One clamp is at an end and other is at 15 cm from other end. The Young's modulus of elasticity and density of the rod are $Y = 1.6 \times 10^{11}$ N/m^2 and $\rho = 2.5$ gm/cc respectively.



(A) the rod will resonate if it is given longitudinal vibrations of frequency 13.33 kHz.
(B) the rod will resonate if it is given longitudinal vibrations of frequency 40 kHz.
(C) during resonance the number of complete loops between clamps may be 6.
(D) the maximum wavelength corresponding to vibrations of the rod is 60 cm.

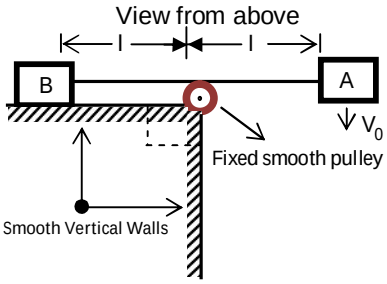
Space for rough work

Paragraph Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Questions Nos. 10 and 11

Two identical blocks are placed on a smooth horizontal surface, connected by a light string of length 2ℓ. String touches a fixed smooth pulley at its mid-point initially.
Shaded parts are two smooth vertical walls. Block A is given a speed V_0 perpendicular to string as shown in diagram. B strikes the pulley and stops.



10. Speed of block B when it hits the pulley is

(A) $\frac{V_0}{2}$ (B) $V_0\frac{\sqrt{3}}{2}$

(C) V_0 (D) $V_0\sqrt{\frac{3}{8}}$

11. Speed of A when it hits the wall is

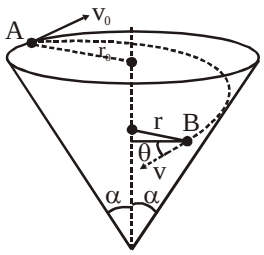
(A) $\frac{V_0}{2}$ (B) $V_0\frac{\sqrt{3}}{2}$

(C) V_0 (D) $V_0\sqrt{\frac{3}{8}}$

Space for rough work

Paragraph for Questions Nos. 12 and 14

A small particle of mass m is given an initial velocity v_0 tangent to the horizontal rim of a smooth cone at a radius r_0 from the vertical centerline as shown at point A. As the particle slides to point B, a vertical distance h below A and a distance r from the vertical centerline, its velocity v makes an angle θ with the horizontal tangent to the cone through B.



12. The value of θ is

(A) $\cos^{-1}\dfrac{v_0 r_0}{\sqrt{v_0^2 r_0^2 + 2gh(r_0 - h\tan\alpha)}}$

(B) $\cos^{-1}\dfrac{v_0 r_0}{\sqrt{v_0^2 r_0^2 + 2gh(r_0 + h\tan\alpha)}}$

(C) $\cos^{-1}\dfrac{v_0 r_0}{\sqrt{v_0^2 r_0^2 - 2gh(r_0 - h\tan\alpha)}}$

(D) $\cos^{-1}\dfrac{v_0 r_0}{r_0\sqrt{v_0^2 + 2gh}}$

13. The speed of particle at point B

(A) $\sqrt{v_0^2 + 2gh}$

(B) $\sqrt{v_0^2 - 2gh}$

(C) $\sqrt{v_0^2 + gh}$

(D) $\sqrt{2v_0^2 + 2gh}$

14. The minimum value of v_0 for which particle will be moving in a horizontal circle of radius r_0.

(A) $\sqrt{\dfrac{2gr_0}{\tan\alpha}}$

(B) $\sqrt{\dfrac{gr_0}{2\tan\alpha}}$

(C) $\sqrt{\dfrac{gr_0}{\tan\alpha}}$

(D) $\sqrt{\dfrac{4gr_0}{\tan\alpha}}$

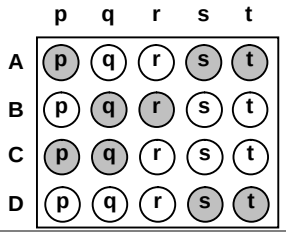
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SECTION – B

(Matrix-Match Type)

This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:



1. For the following statements, except gravity and contact force between the contact surfaces, no other force is acting on the body.

Column A		Column B	
(A)	When a sphere is in pure–rolling on a fixed horizontal surface.	(p)	Upward direction
(B)	When a cylinder is in pure rolling on a fixed inclined plane in upward direction then friction force acts in	(q)	$v_{cm} > R\,\omega$
(C)	When a cylinder is in pure rolling down a fixed incline plane, friction force acts is	(r)	$v_{cm} < R\omega$
(D)	When a sphere of radius R is rolling with slipping on a fixed horizontal surface, the relation between v_{cm} and ω is	(s)	No frictional force acts.
		(t)	none of these

2. Match the column I with column II
In simple harmonic motion: x = 1.0 sin [12πt] and mass of particle executing SHM, m = 1/4 kg

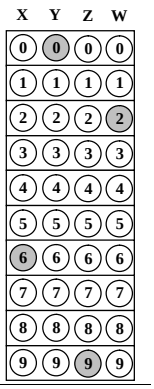
Column A		Column B	
(A)	Frequency with which kinetic energy oscillates = …	(p)	$\frac{1}{12}$
(B)	Speed of particle is maximum at time = ….	(q)	$18\pi^2$
(C)	Maximum potential energy = …	(r)	12
(D)	Force constant k = …	(s)	$36\pi^2$
		(t)	none of these

Space for rough work

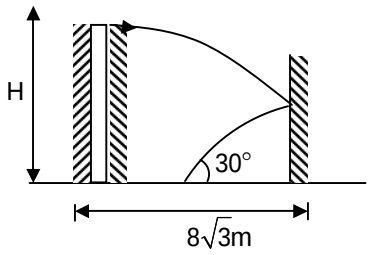
SECTION – C

Integer Answer Type

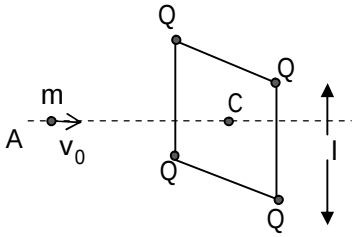
This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. A ball is thrown horizontally from the top of a tower of unknown height. Ball strikes a vertical wall whose plane is normal to the plane of motion of ball. Collision is elastic and ball falls on ground exactly at the midpoint between the tower and the wall. Ball strikes the ground at an angle of the 30° with horizontal. Find the height of the tower (in meter).

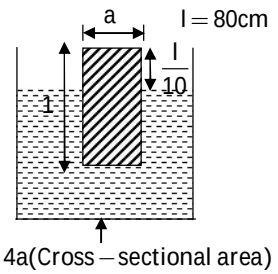


2. Four identical point charges are fixed at the four corners of a square of a side length l. Another charged particle of mass m and charge +q is projected towards centre of square from a large distance along the line perpendicular to plane of square. The minimum value of initial velocity v_0 required to cross the square is?
(m = 1 gm, l = $4\sqrt{2}$ m, Q = 1 μc, q = 0.5 μc)

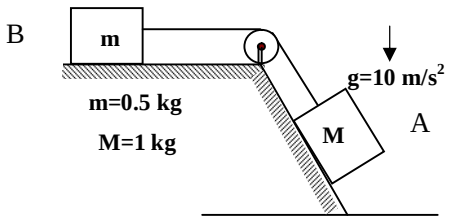


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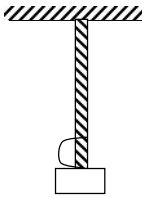
3. A uniform vertical cylinder of cross-sectional area 'a' floats, 90% submerged, in an unknown liquid inside a tank with cross-sectional area four times that of cylinder. When cylinder is pushed down gently and released it performs SHM. The maximum possible amplitude (in cm) for this SHM is



4. Horizontal surface below block B is rough but inclined surface below A is smooth as shown in diagram. When A is given a small impulse down the incline, blocks are found to slide with constant speed. T is tension in the string and μ is coefficient of friction between block B and horizontal surface. The value of $\frac{T}{\mu}$ is (in Newton)



5. A rope whose mass is not negligible supports a block of mass three times that of rope. A pulse is generated in string near lower end, the pulse moves up with acceleration in(m/s^2).



Space for rough work

Chemistry

PART – II

SECTION – A

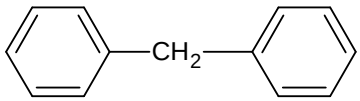
Straight Objective Type

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Ordinary hydrogen at room temperature is a mixture of
   (A) 75% of O – hydrogen + 25 % of p – hydrogen
   (B) 25% of O – hydrogen + 75 % of p – hydrogen
   (C) 50% of O – hydrogen + 50 % of p – hydrogen
   (D) 1% of O – hydrogen + 99 % of p – hydrogen

2. The enolic form of acetone contains
   (A) 8 σ - bonds, 2π - bonds and 1 lone pairs
   (B) 9 σ - bonds, 1π - bonds and 2 lone pairs
   (C) 9 σ - bonds, 2π - bonds and 1 lone pairs
   (D) 10 σ - bonds, 1π - bonds and 1 lone pairs

3. The molecular formula of diphenyl methane



   is $C_{13}H_{12}$.
   How many structural isomers are possible, when one of the hydrogen is replaced by a chlorine atom?
   (A) 8
   (B) 7
   (C) 6
   (D) 4

4. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4. In the nitrating mixture HNO_3 act as a
   (A) Base
   (B) Acid
   (C) reducing agent
   (D) catalyst

5. The degree of dissociation α of the reaction:
   $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ can be related to K_p as, where P is total pressure at equilibrium.
   (A) $\alpha = \dfrac{K_p/P}{4+K_p/P}$
   (B) $\alpha = \dfrac{K_p}{4+K_p}$
   (C) $\alpha = \left[\dfrac{K_p/P}{4+K_p/P}\right]^{1/2}$
   (D) $\alpha = \dfrac{K_p}{4+K_p}$

6. The solubility of A_2X_3 is Y mol dm^{-3}. if solubility product is
   (A) $6Y^4$
   (B) $64\ Y^4$
   (C) $36\ Y^5$
   (D) $108\ Y^5$

Space for rough work

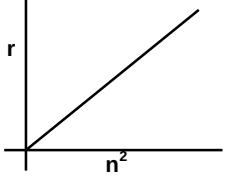
Multiple Correct Choice Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

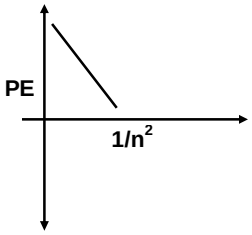
7. Which of the following is/are correct?
   (A) α - rays are more penetrating than β - rays
   (B) α - rays have greater ionizing power than β -rays
   (C) β - particles are not present in the nucleus yet they are emitted from the nucleus
   (D) γ - rays are not emitted simultaneously with α and β - rays

8. Select the correctly graph presented if
   v = velocity of electron in Bohr's orbit
   r = radius of Bohr's orbit
   U = potential energy
   T = kinetic energy of electron in Bohr's orbit

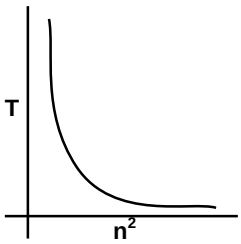
(A)



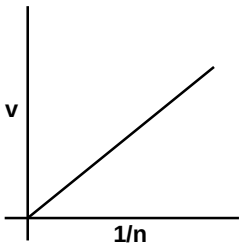
(B)



(C)



(D)



9. Which of the following name reactions may be used in the preparation of alkanes?
   (A) Corey – house synthesis
   (B) Wurtz reaction
   (C) Aldol reaction
   (D) Kolbe electrolysis method

Space for rough work

Paragraph Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 to 11

The concentration of hydrogen ion is a measure of acidity or alkalinity of a solution.
For mono basic acids

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$\alpha = \sqrt{\frac{K_a}{C}}$

Where K_a = dissociation constant of acid
C = molarity of acid

$[H^+] = C\alpha$

$\left[H^+\right] = \frac{K_a}{\alpha}$

$\left[H^+\right] = \sqrt{CK_a}$

pH of a weak acid can be calculated using $[H^+]$ by any of above methods.

$\left[H^+\right]_{total} = \sqrt{C_1K_{a_1} + C_2K_{a_2}}$ [where two weak acids are mixed]

$[H^+]$ of polyprotic acid [weak H_3A] has dissociation constant $K_{a_1} >> K_{a_2} >> K_{a_3}$ will be contributed by first dissociation at the most.

10. What will be value of $[H^+]$ of 10^{-6} M CH_3COOH? [$K_a = 1.8 \times 10^{-5}$]
(A) 4.2×10^{-6}M (B) 10^{-4}M
(C) 9.5×10^{-7}M (D) 10^{-10}M

11. The pH of 0.01 M HCOOH ($K_a = 1.4 \times 10^{-4}$) will be
(A) 2.926 (B) 3.296
(C) 4.926 (D) 5.926

Space for rough work

Paragraph for Question Nos. 12 to 14

Resonance concept is purely the critical and base on valence bond theory. According to this theory certain molecules showing at least one π bond can be represented by two or more than possible structures but none of them is real one. The real structure is hybrid of all these possible structure. This phenomenon is known as resonance.

12. In which resonance phenomenon is not possible?
(A) SO_4^{2-} (B) ClO_4^-
(C) CO_3^{2-} (D) CH_4

13. Bond order of Cl – O bond in ClO_4^- is
(A) 3 (B) 2
(C) 1.5 (D) 1.75

14. The carbon oxygen bond length order is as follows:
(A) $CO < CO_2 < CO_3^{2-}$ (B) $CO_2 > CO > CO_3^{2-}$
(C) $CO_3^{2-} > CO > CO_2$ (D) $CO_2 > CO_3^{2-} > CO$

Space for rough work

SECTION – B

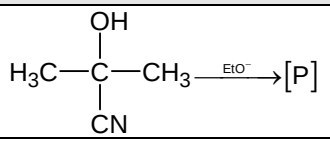
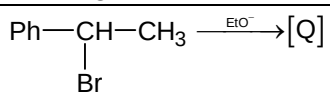
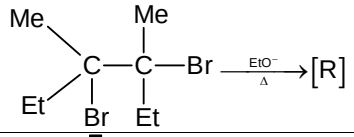
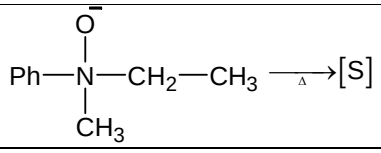
(Matrix-Match Type)

This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:

	p	q	r	s	t
A	●	○	○	●	○
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	○

1. Match the column – I with chemicals of that in valence in the process:

Column – I		Column – II	
(A)		(p)	E_1
(B)		(q)	E_2
(C)		(r)	E_{1CB}
(D)		(s)	E_i (Intramolecular elimination)
		(t)	Hoffmann elimination

2. All the compounds listed in Column-I react with water and in Column-II there product Column-II

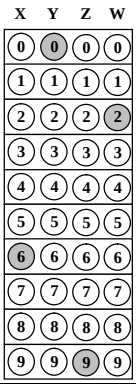
Column – I		Column – II	
(A)	$(CH_3)_2SiCl_2$	(p)	Hydrogen halide formation
(B)	XeF_4	(q)	Redox reaction
(C)	Cl_2	(r)	Reacts with glass
(D)	VCl_5	(s)	Polymerization
		(t)	O_2 formation

Space for rough work

SECTION – C

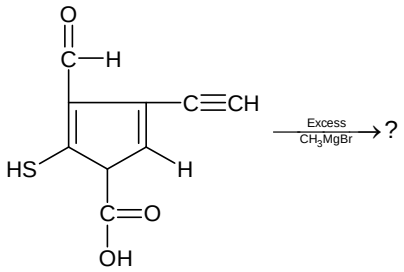
Integer Answer Type

This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. What is the ratio of mole of $Mg(OH)_2$ and $Al(OH)_3$ in one litre in their saturated solution?
   K_{sp} of $Mg(OH)_2 = 4 \times 10^{-12}$, $Al(OH)_3 = 1 \times 10^{-33}$
   Gives answer by multiply it with 10^{-17}.

2. Find energy level of Li^{2+} having velocity 2.18×10^6 m/sec.

3. 

   The number of CH_4 gas formed in the reaction are.

4. Most probable speed of a gas at 150 K is equal to the root mean square speed of O_2 gas at 400 K. Molecular mass of the gas is.

5. Total number of isomers (including stereoisomers) obtained by monochlorination of 2, 2, 4 – trimethylpentane is/are.

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If a, b, c, d $\in$ R then the equation $(x^2 + ax – 3b)(x^2 – cx + b)(x^2 – dx + 2b) = 0$ has
   (A) 6 real roots (B) 3 real roots
   (C) 4 real roots (D) at least 2 real roots

2. The number of values of λ for which $[x^2 –(\lambda – 2)x + \lambda^2][x^2 + \lambda x + (2\lambda – 1)]$ is a perfect square is
   (A) 1 (B) 2
   (C) 0 (D) none of these.

3. If the equation of the family of ellipse is
   $\frac{x^2}{\cos^2\theta} + \frac{y^2}{\sin^2\theta} = 1\left(0 < \theta < \frac{\pi}{4}\right)$
   then the locus of the extremities of the latus rectum is
   (A) $2y(1 – x^2) = 1 + x^2$ (B) $2y^2(1 + x^2) = (1 – x^2)^2$
   (C) $y^2 = (1 – x^2)^2$ (D) $2y^2(1 – x^2) = (1 + x^2)^2$.

4. If PQ and QR are two focal chords of an ellipse and the eccentric angles of P, Q, R are 2α, 2β, 2γ respectively. Then $\tan\alpha$. $\tan\gamma$ is equal to
   (A) $\cot^2\beta$ (B) $\cot^2\alpha$
   (C) $2\cot\alpha$ (D) none of these.

5. A focal chord of parabola $y^2 = 4x$ is inclined at an angle of $\frac{\pi}{4}$ with positive x-direction, then the slope of normal drawn at the ends of chord will satisfy the equation
   (A) $m^2 – 2m + 1 = 0$ (B) $m^2 + 2m – 1 = 0$
   (C) $m^2 – 1 = 0$ (D) $m^2 + 2m – 2 = 0$

6. The equation of a circle is $x^2 + y^2 – \lambda ax – ay – a^2(\lambda^2 + 1) = 0$. If two chords, each bisected by the x-axis can be drawn to the circle from the point $(a(\lambda – 1), a(\lambda + 1))$, which lies on the circle then λ satieties
   (A) $\lambda^2 – 4\lambda – 12 > 0$ (B) $\lambda^2 – 4\lambda – 12 < 0$
   (C) $\lambda^2 – 12\lambda – 4 > 0$ (D) $\lambda^2 – 12\lambda – 4 < 0$.

Space for rough work

Multiple Correct Choice Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. The points A(0, 0), B($\cos\alpha$, $\sin\alpha$) and C($\cos\beta$, $\sin\beta$) are the vertices of a right angled triangle if

(A) $\sin\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (B) $\cos\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$

(C) $\cos\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (D) $\sin\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$.

8. Two straight lines u = 0 and v = 0 passes through the origin and angle between them is $\tan^{-1}\left(\frac{7}{9}\right)$. If the ratio of the slope of v = 0 and u = 0 is $\frac{9}{2}$, then their equations are

(A) y = 3x & 3y = 2x (B) 2y = 3x & 3y = x

(C) y + 3x = 0 & 3y + 2x = 0 (D) 2y + 3x = 0 & 3y + x = 0.

9. If P, Q and R are three co-normal points on the parabola $y^2 = 4ax$. Then the correct statement(s) is/are

(A) algebraic sum of the slopes of the normals at P, Q and R vanishes
(B) algebraic sum of the ordinates of points P, Q and R vanishes
(C) centroid of the triangle PQR lies on the axis of the parabola
(D) circle circumscribing the triangle PQR passes through the vertex of the parabola

Space for rough work

Paragraph Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Questions 10 and 11

Read the following write up carefully and answer the following questions:

Let $f(x) = x^2 + b_1x + c_1$, $g(x) = x^2 + b_2x + c_2$. Real roots of $f(x) = 0$ be α, β and real roots of $g(x) = 0$ be $\alpha + \delta$, $\beta + \delta$. Least value of $f(x)$ be $-\frac{1}{4}$, least value of $g(x)$ occurs at $x = \frac{7}{2}$

10. The least value of g(x) is

(A) -1 (B) $-\frac{1}{2}$

(C) $-\frac{1}{4}$ (D) $-\frac{1}{3}$

11. The value of b_2 is

(A) 6 (B) -7

(C) 8 (D) 0.

Paragraph for Questions 12 and 14

Read the following write up carefully and answer the following questions:

Tangents PA and PB are drawn to the circle $(x-4)^2 + (y-5)^2 = 4$, from the point P on the curve $y = \sin x$, where A and B lie on the circle. Consider the function $y = f(x)$ represented by the locus of the centre of the circum-circle of triangle PAB, then answer the following questions.

12. Range of $y = f(x)$ is

(A) $[-2, 1]$ (B) $[-1, 4]$

(C) $[0, 2]$ (D) $[2, 3]$

13. Period of $y = f(x)$ is

(A) 2π (B) 3π

(C) π (D) not defined.

14. Which of the following is true

(A) $f(x) = 4$, has real roots (B) $f(x) = 1$, has real roots

(C) Range of $y = f^{-1}(x)$ is $\left[-\frac{\pi}{4}+2, \frac{\pi}{4}+2\right]$ (D) none of these.

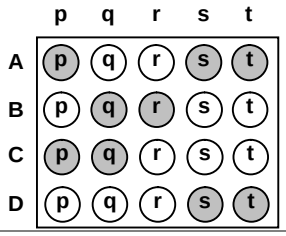
Space for rough work

SECTION – B

(Matrix-Match Type)

This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:



1. Match the following Column-I with Column-II

Column – I	Column – II
(A) Let N be the no. of solutions of the equation $\|x^2 - x - 6\| = x + 2$, then the value of N is	(p) –3
(B) If $x^2 + \lambda x + 1 = 0$ and $(b - c)x^2 + (c - a)x + (a - b) = 0$ have both the roots common, then $[\lambda - 1]$ is (where [.] denotes the greatest integer function)	(q) – 2
(C) If, $-3 < \frac{x^2 - \lambda x - 2}{x^2 + x + 1} < 2$, for $x \in R$, then $[\lambda]$ is can be where [.] denotes the greatest integer function	(r) –1
(D) the equation $x^3 - 6x^2 + 9x + \lambda = 0$ have exactly one root in (1, 3), then $[\lambda + 1]$ is (where [.] denotes the greatest integer function)	(s) 3
	(t) 0

2. Match the following Column-I with Column-II

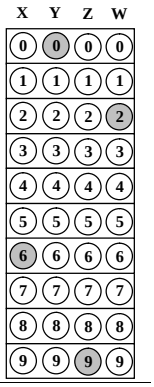
Column – I	Column – II
(A) Lines $x - 2y - 6 = 0$, $3x + y - 4 = 0$ and $\lambda x + 4y + \lambda^2 = 0$, are concurrent, then the value of λ is	(p) 2
(B) The points $(\lambda + 1, 1)$,$(2\lambda + 1, 3)$ and $(2\lambda + 2, 2\lambda)$ are collinear, then the value of λ is	(q) 4
(C) If line $x + y - 1 - \lambda = 0$, passing through the intersection of $x - y + 1 = 0$ and $3x + y - 5 = 0$, is perpendicular to one of them then the value of λ is	(r) – ½
(D) If line $y - x - 1 + \lambda = 0$ is equally inclined to axes and equidistant from the points (1, – 2) and (3, 4), then λ is	(s) – 4
	(t) 3

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. The area of a triangle whose vertices are (b, c), (c, a) and (a, b) is Δ and the area of the triangle whose vertices are $(ac - b^2, ab - c^2)$, $(ab - c^2, bc - a^2)$ and $(bc, -a^2, ac - b^2)$ is Δ'. If a, b, c represent the sides of a triangle of perimeter 16. Then $\frac{\Delta'}{256\Delta}$ is equal to _____

2. If $ax + by - 5 = 0$ is the equation of the chord of the circle $(x - 3)^2 + (y - 4)^2 = 4$, which passes through (2, 3) and at the greatest distance from the centre of the circle, then $|a + b|$ is ____

3. The reflection of the point $(t - 1, 2t + 2)$ in a line is $(2t + 1, t)$ then the line has slope equals to ___

4. Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and contained in it, is _____

5. The shortest distance between parabolas $y^2 = 4x$ and $y^2 = 2x - 6$ is d then d^2 is _____

Space for rough work